

PRODUCT SUBMISSION DOCUMENT

RailConnect

Railway AI Project — Smart Train Search, Booking & AI Travel Assistant

Version 1.0

Date: July 3, 2026

Owner: KIRTI ANURAG

Status: Active Development

Confidential — Internal Use Only

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1. Product Overview

One-liner: RailConnect (Railway AI Project) is a modern web application that simplifies railway travel by offering train searches, ticket booking, PNR status checking, and an integrated AI assistant powered by Google Gemini for intelligent travel insights.

Field	Detail
Product Name	RailConnect (Railway AI Project)
Owner / Team	KIRTI ANURAG
Date	July 3, 2026
Version	1.0
Primary Language	JavaScript (React + Node.js)
License	ISC

Description: RailConnect is a full-stack web application designed for railway travelers. It features a React-based frontend styled with Tailwind CSS, backed by a Node.js + Express server and MongoDB. Key functionalities include user authentication, train search, ticket booking, PNR status tracking, and a smart AI chatbot powered by Gemini to provide real-time travel insights and support.

2. Objectives and Goals

- Simplify Train Discovery — Allow users to search for trains between stations seamlessly.
- Facilitate Easy Booking — Provide a straightforward interface for checking seat availability, fare calculation, and booking tickets.
- Enhance User Support with AI — Integrate a Gemini-powered AI chatbot to answer travel queries, provide schedule reliability insights, and assist users 24/7.
- Provide Real-Time Tracking — Enable users to check their PNR status instantly.
- Secure User Data — Ensure authenticated access via JWT and securely store user data in MongoDB.
- Deliver a Premium UI/UX — Offer a modern, responsive, and aesthetically pleasing interface (dark theme, glassmorphism) to improve user engagement.

3. Target Users

User Persona	Type	Description
Commuters / Travelers	Primary	Individuals looking to book train tickets, check PNR statuses, and manage their bookings.
First-Time Travelers	Secondary	Users who need guidance on train travel and can benefit from the AI Assistant.
Railway Enthusiasts	Secondary	People interested in checking train schedules, routes, and seat availability.

4. Actual Problem Statement

Core Problem: Traditional railway booking platforms often suffer from cluttered interfaces, lack of intuitive AI support for immediate query resolution, and fragmented user experiences across searching, booking, and tracking.

Specific Pain Points

- Overwhelming UI — Legacy systems are often dense and hard to navigate for new users.
- Lack of Instant Support — Users struggle to find quick answers about train reliability, food quality, or general FAQs without calling customer care.
- Scattered Information — Checking PNR, calculating fares, and viewing schedules often require navigating multiple complex menus.

5. Proposed Solution

RailConnect addresses these issues through a unified, AI-enhanced platform with a three-layer architecture:

- Layer 1 — Authentication & Data: MongoDB securely stores user credentials (bcrypt hashed) and booking history. JWT manages sessions.
- Layer 2 — Backend Engine: A Node.js + Express backend handles train queries, PNR lookups, and booking logic. It also interfaces with the Google Gemini API for intelligent chat support.
- Layer 3 — Rich Frontend: A React + Vite SPA with Tailwind CSS delivers a fast, responsive, and visually stunning user interface, including a dedicated Dashboard and AI chat console.

Why This Works

- Clean, modern aesthetic reduces cognitive load.
- The integrated Gemini AI provides instant, conversational help.
- Unified dashboard brings search, booking, and PNR tracking into one cohesive workflow.

6. Features

6.1 F1 — Secure Authentication

- User registration and login with form validation.
- Passwords hashed using bcrypt before storage.
- JWT issued on successful login to authorize protected API routes.

6.2 F2 — Train Search & Seat Availability

- Search trains by source and destination stations.
- View real-time seat availability and class-wise fare calculations.

6.3 F3 — Ticket Booking & Management

- Streamlined ticket booking process with passenger detail entry.
- Mock payment gateway integration.
- Dedicated Booking History dashboard for users to review past trips.

6.4 F4 — PNR Status Checker

- Instantly retrieve the status of a booked ticket using its PNR number.

6.5 F5 — Gemini AI Assistant

- Conversational chatbot answering railway-related queries.
- Graceful fallback responses during API rate limits.
- Provides structured insights on schedules and catering.

6.6 F6 — Premium Aesthetic UI

- Responsive React frontend with Tailwind CSS.
- Features crossfade slideshows, frosted glass overlays, and dynamic hover effects.

7. Use Flow

7.1 Primary Flow — Booking a Ticket

- Register/Login — User registers or logs into their account.
- Dashboard — User arrives at the main Dashboard.
- Search — User navigates to Search Trains, entering source and destination.
- Select — User selects a train, checking Seat Availability and Fare.
- Book — User proceeds to Book Ticket and enters passenger details.
- Payment — User completes the payment process.
- Ticket — User views the final Ticket, which is saved in their Booking History.

7.2 Secondary Flow — AI Assistance

- Encounter Issue — User has a question or needs travel insights.
- Open Chat — User opens the AI Assistant interface.
- Query — User asks a question (e.g., "insight summary for train X").
- Response — User receives a structured, helpful response from the Gemini API.

8. Functional Requirements

ID	Feature	Requirement	Priority
FR-01	User Registration	System shall allow new users to register with credentials.	High
FR-02	User Login	System shall validate credentials and issue a signed JWT.	High
FR-03	Train Search	System shall fetch and display available trains between stations.	High
FR-04	Ticket Booking	System shall process booking requests and store them in DB.	High
FR-05	PNR Status	System shall return ticket details given a valid PNR.	High
FR-06	AI Chat	System shall send user prompts to the Gemini API and display responses.	Medium
FR-07	Booking History	System shall display all past bookings for the logged-in user.	Medium

9. Non-Functional Requirements

9.1 Performance

- Page Load Time: < 2 seconds on a modern browser (Vite + React SPA).
- API Latency: Backend APIs must respond efficiently to avoid user wait times.

9.2 Security

- All passwords must be stored as bcrypt hashes.
- JWT tokens must be signed with a server-side secret and have an expiry.
- Protected routes must reject requests if JWT is absent or invalid.
- API Keys (like Gemini) must be stored in .env and never exposed to the frontend.

9.3 Reliability

- Graceful error handling for AI quota limits with fallback responses.
- Reliable MongoDB connection with robust error logging.

9.4 Usability

- Intuitive dashboard allowing non-technical users to navigate easily.
- Mobile-responsive layout.

9.5 Maintainability

- Clear separation of frontend (React) and backend (Node/Express).
- Modular routing and controllers in the backend.

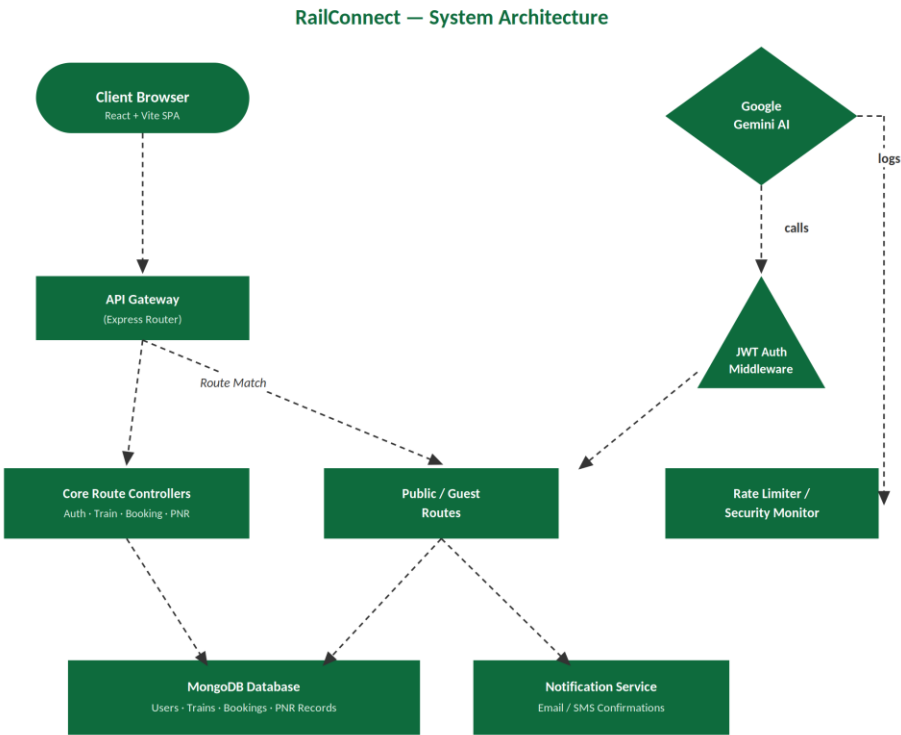
9.6 Portability

- Must run on any OS with Node.js installed.
- Deployable to standard cloud platforms (e.g., Vercel for frontend, Render/Railway for backend).

10. Tech Stack

Layer	Technology	Purpose
Frontend Framework	React + Vite	SPA with fast HMR development server
Styling	Tailwind CSS	Utility-first dark theme UI
Backend Runtime	Node.js	JavaScript server runtime
Backend Framework	Express.js	HTTP routing and middleware
Database	MongoDB + Mongoose	NoSQL database for users, bookings, trains
Auth — Hashing	bcryptjs	Password hashing with salt rounds
Auth — Sessions	JWT	Stateless token-based authentication
AI Integration	@google/generative-ai	Natural language processing for assistant
HTTP Client	Axios	Making API requests from FE to BE

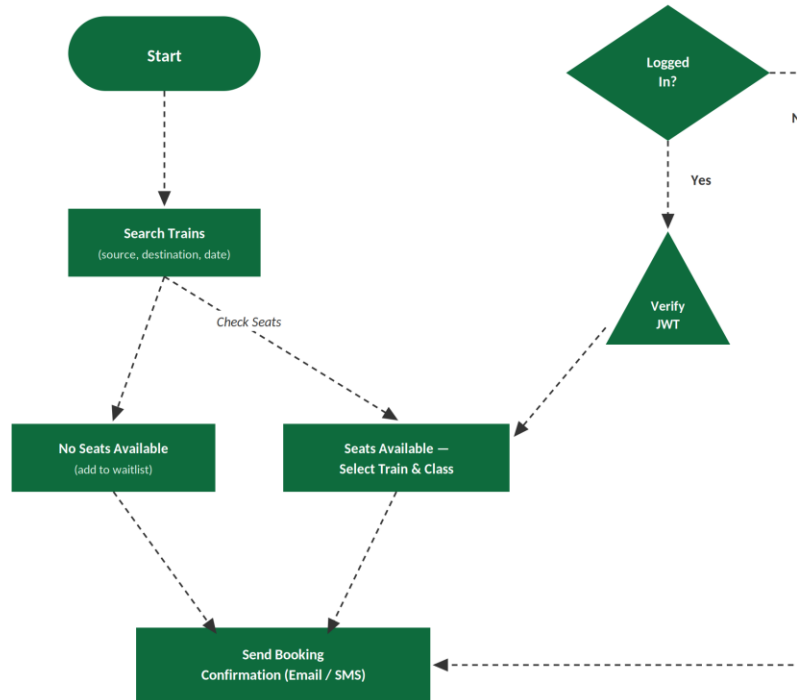
Architecture Diagram (Logical)



The client (React SPA) calls the Express API Gateway over REST/JSON. Authenticated requests pass through JWT middleware to Google Gemini for AI insights, while the Core Route Controllers handle booking, train, and PNR logic against MongoDB, with a separate public-route path for guest access.

Data Flow Diagram (DFD) — Booking Process

RailConnect — Booking Data Flow Diagram (DFD)



All connectors represent data / control flow between processes and decision points.

A user starts by searching trains; seat availability determines whether they are waitlisted or proceed to select a train and class. In parallel, login status is checked — authenticated sessions are verified via JWT before merging back into the booking path — and every path converges on a booking confirmation sent by email or SMS.

11. Risk Assessment

#	Risk	Likelihood	Impact	Severity	Mitigation
R1	Gemini API Rate Limits — Free tier quota exhaustion.	High	Medium	Medium	Implement graceful fallback responses in the controller to ensure uninterrupted UX.
R2	DB Connection Failure — MongoDB Atlas goes down or network issues.	Low	High	High	Implement retry logic and robust error messages.
R3	Security Misconfiguration — Exposed JWT secret or Gemini API keys.	Low	High	High	Strict use of .env files, .gitignore, and environment variables in production.
R4	Concurrency Issues — Multiple users attempting to book the same seat.	Medium	High	High	Implement database transactions or seat locking mechanisms in the future.

Risk Priority Matrix

	Low Impact	Medium Impact	High Impact
High Likelihood	—	R1	—
Medium Likelihood	—	—	R4
Low Likelihood	—	—	R2, R3

12. Screenshots

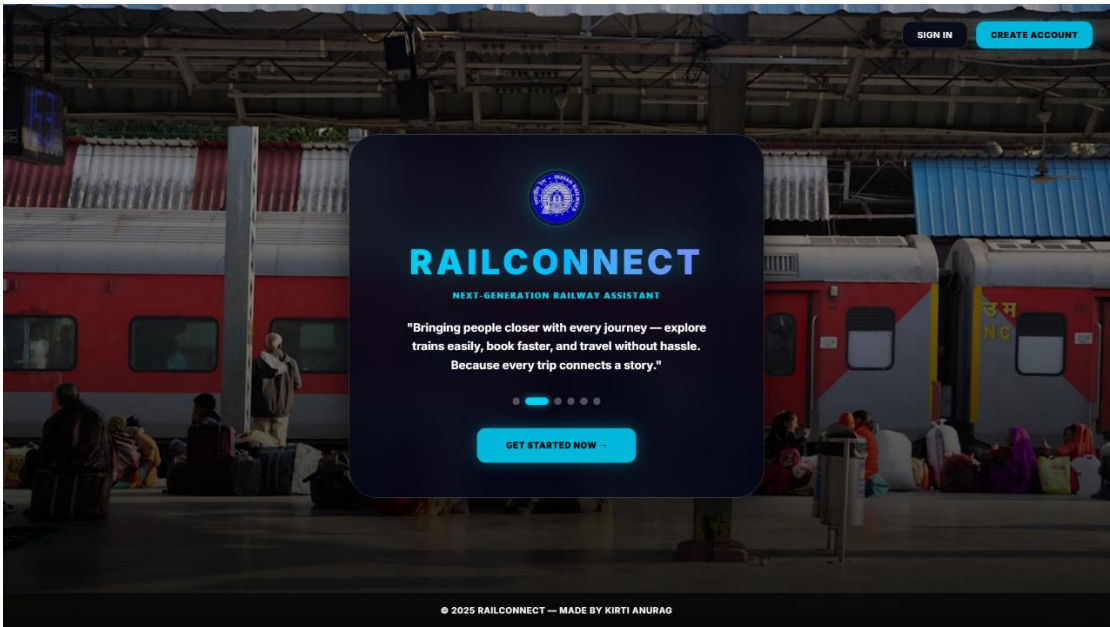


Fig. 1.0 Welcome Page

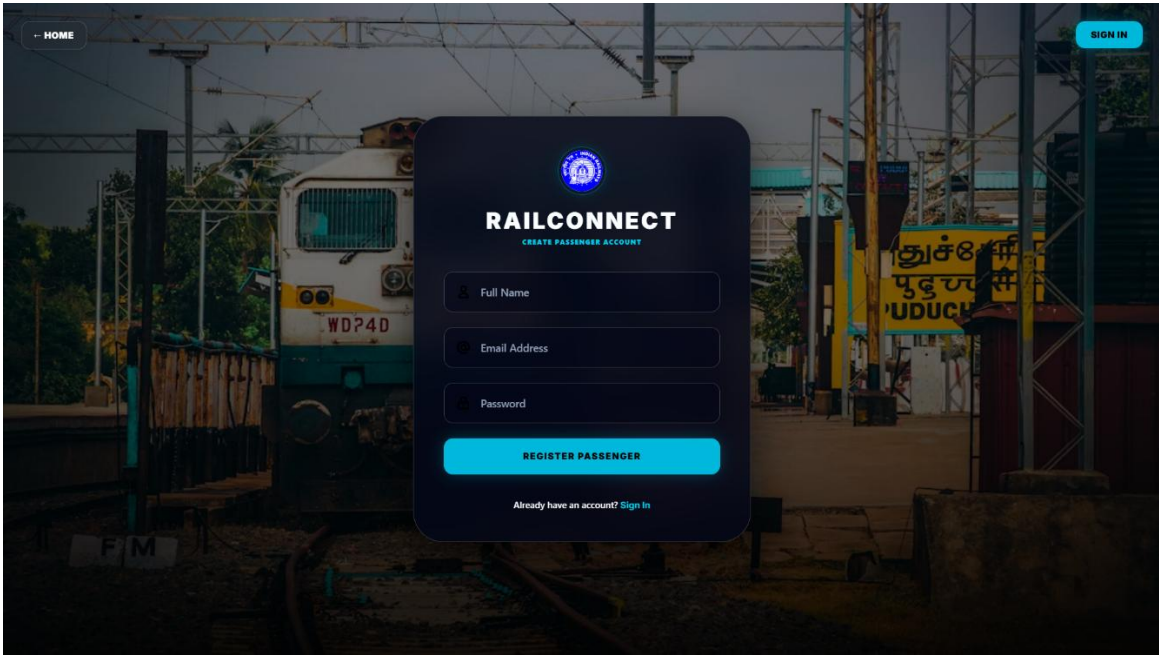


Fig. 1.1 Sign In Page

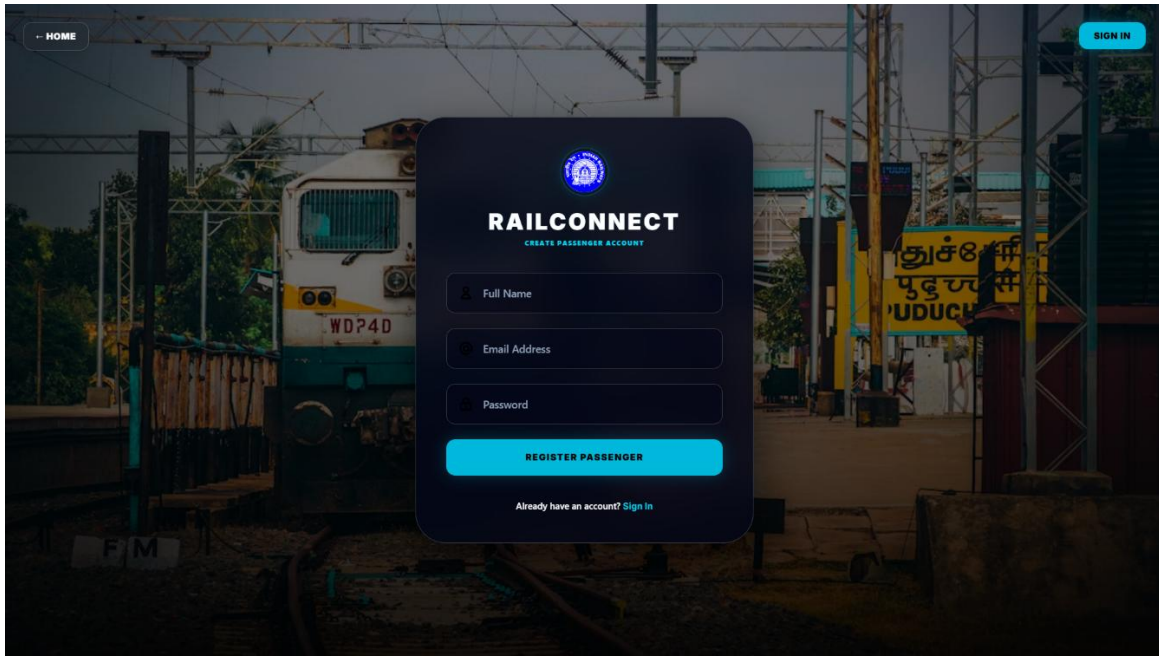


Fig. 1.2 Create new account Page

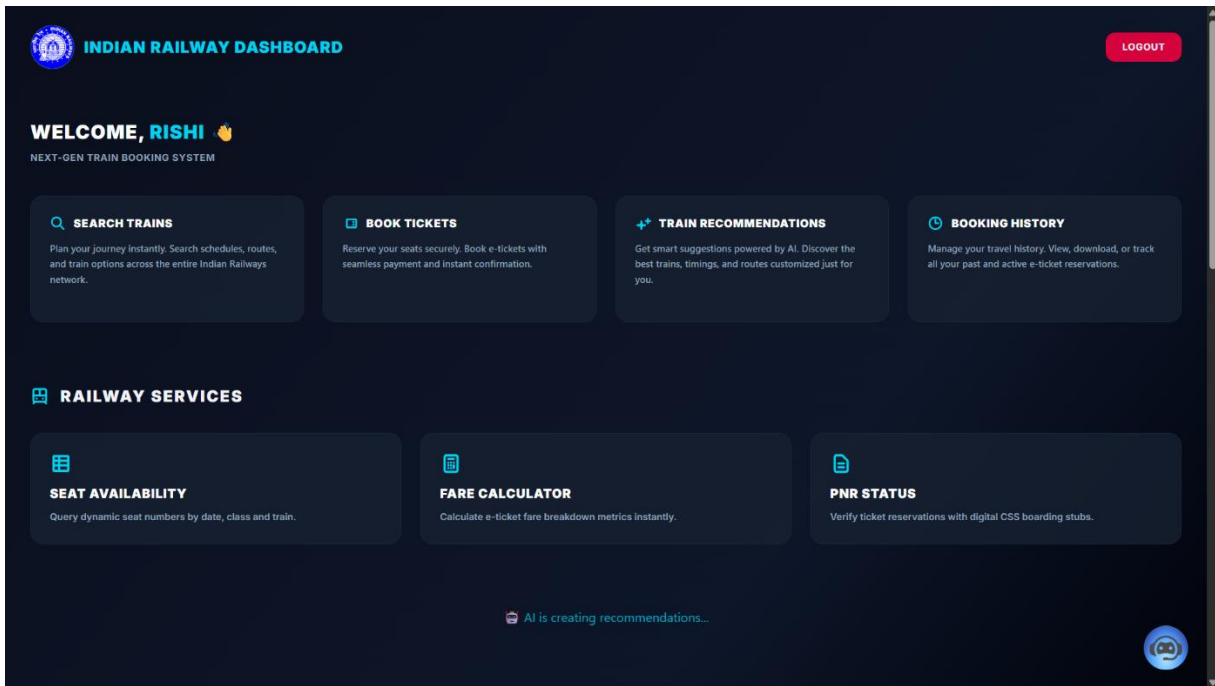


Fig. 1.3 Dashboard

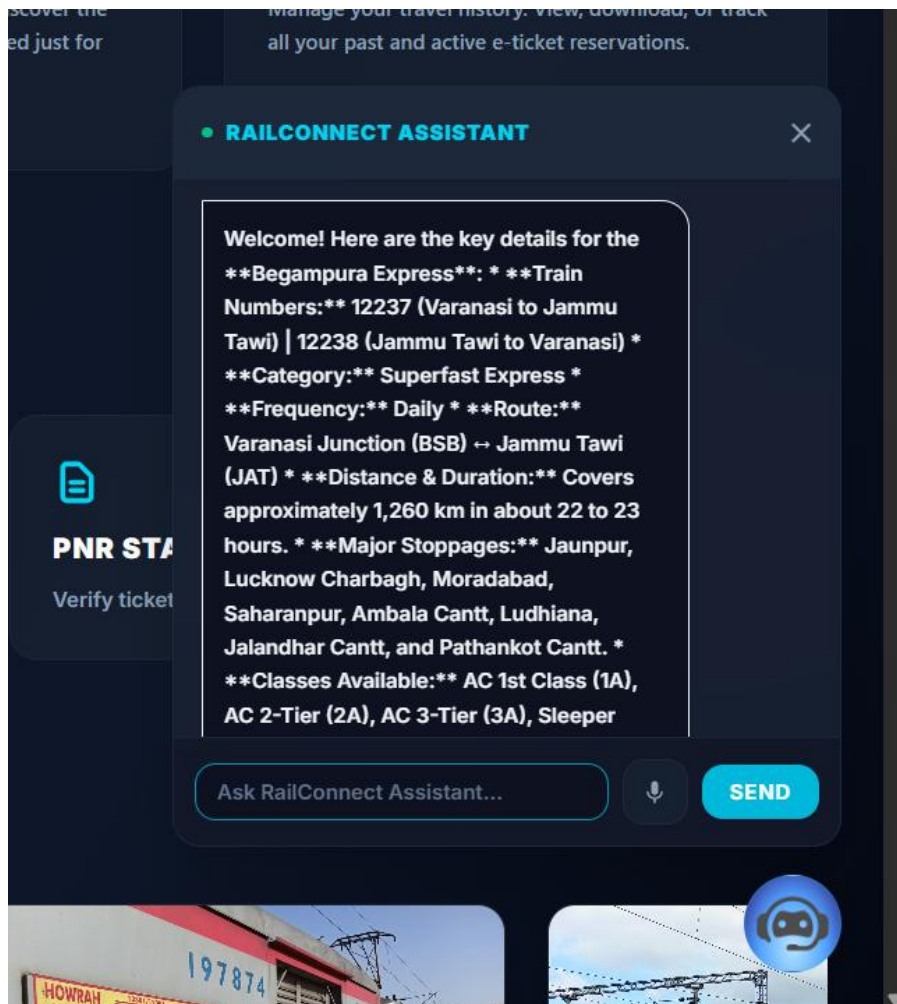


Fig. 1.4 Talking with Chatbot

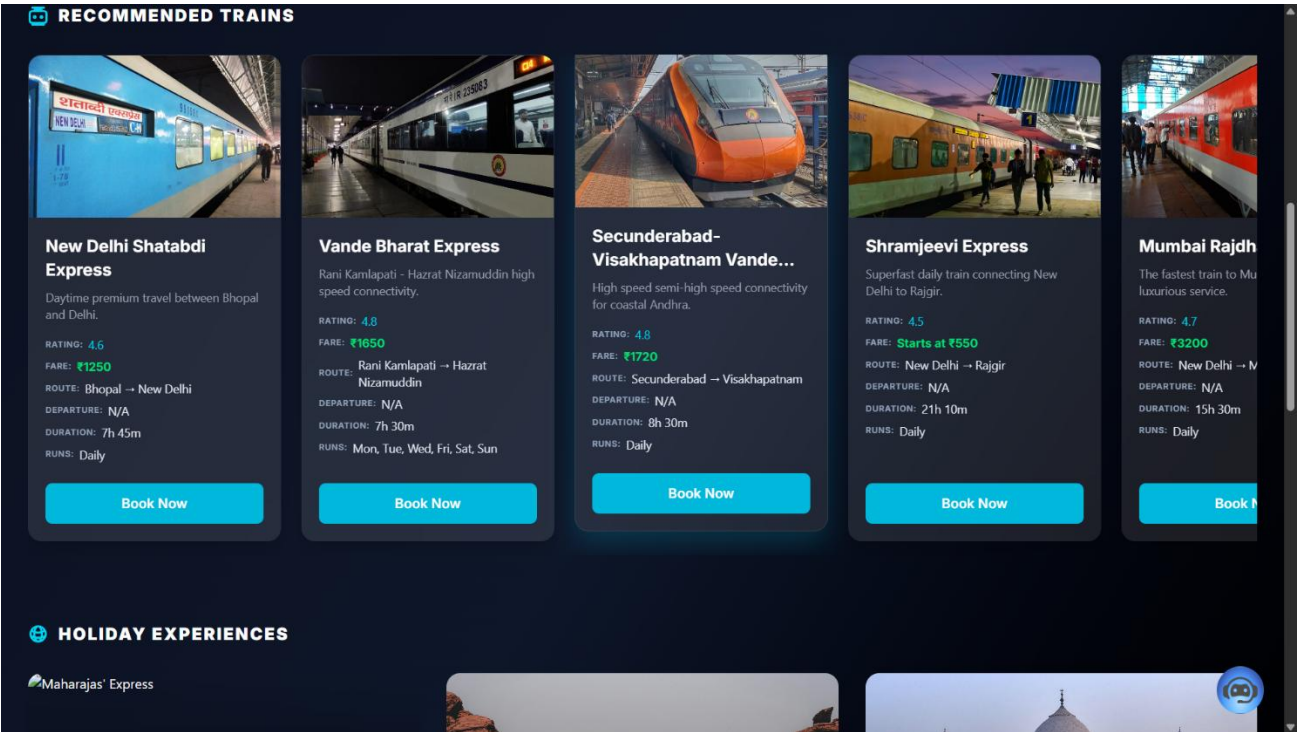


Fig. 1.5 Dashboard

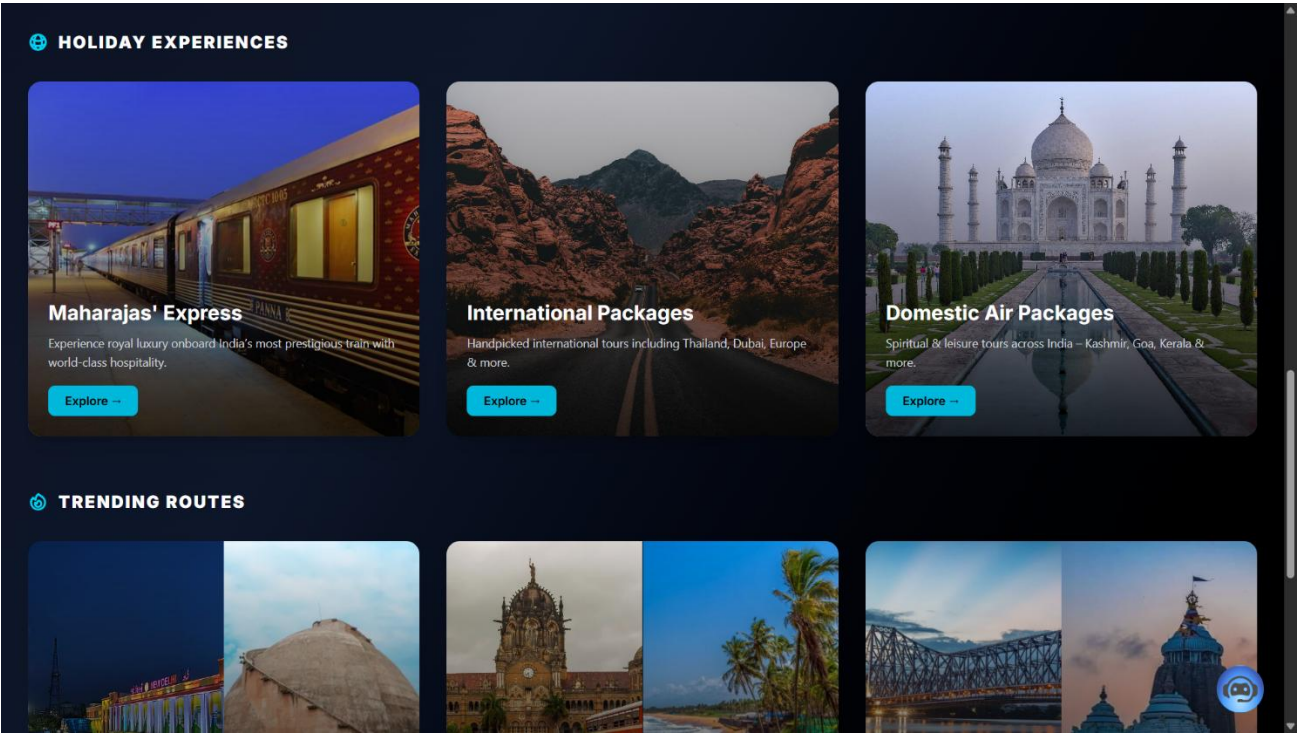


Fig. 1.6 Dashboard

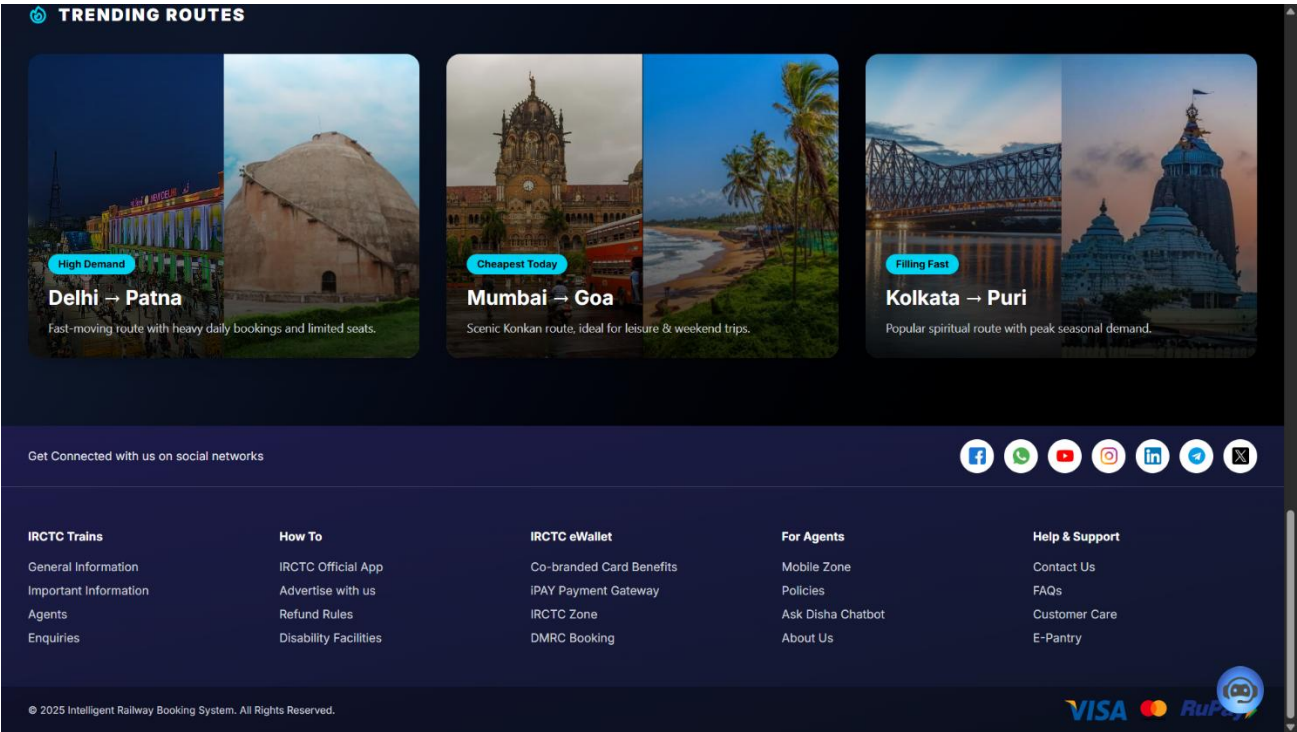


Fig. 1.7 Dashboard

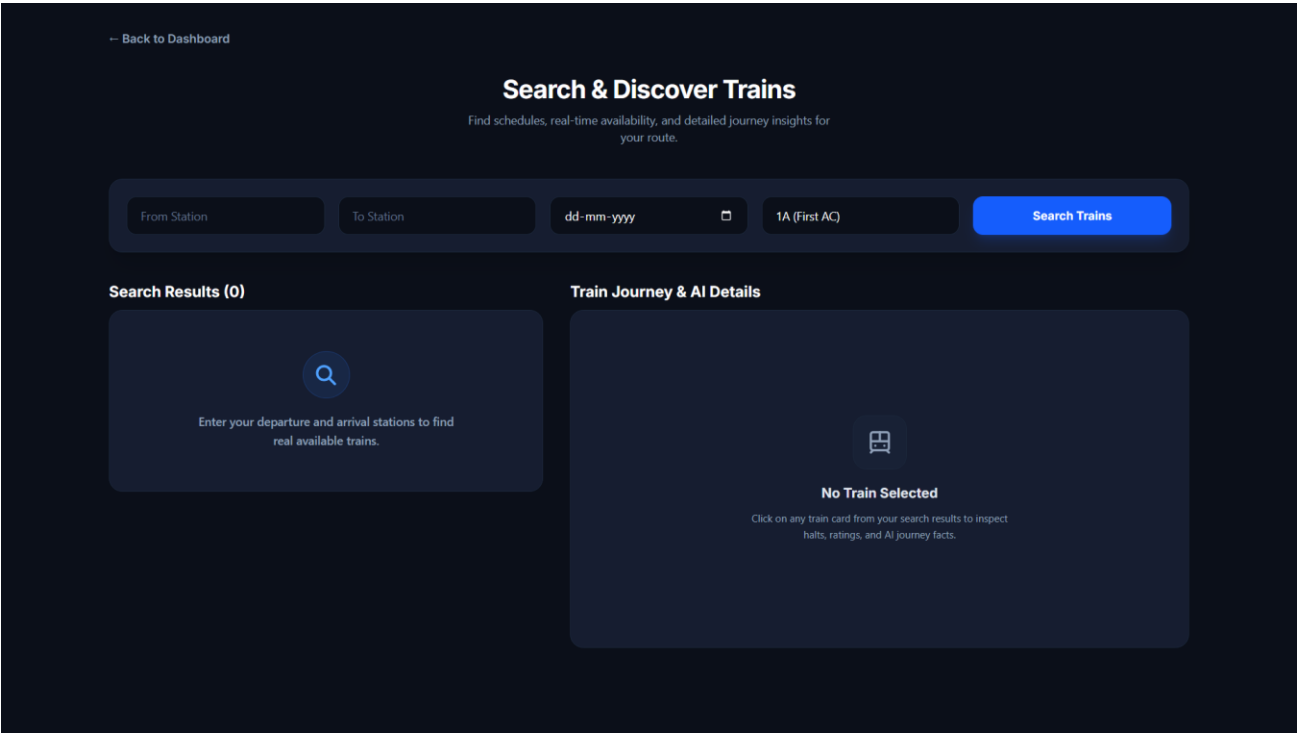


Fig. 1.8 Train Search Page

— Back to Dashboard

Search & Discover Trains

Find schedules, real-time availability, and detailed journey insights for your route.

Search Results (5)

Vande Bharat Express
No: 22436 • Mon, Tue, Wed, Fri, Sat, Sun
Delhi → Varanasi
DURATION: 8H 00M
₹1750

Shiv Ganga Express
No: 12560 • Daily
Delhi → Varanasi
DURATION: 9H 50M
₹2720

Mahamana Express
No: 22418 • Tue, Thu, Sat
Delhi → Varanasi
DURATION: 13H 50M
₹2900

Train Journey & AI Details

★ 4.9 RATED SERVICE

Vande Bharat Express
Train No: 22436 • Speed-link

DURATION: 8H 00M

Features high-speed semi-executive comfort, bio-toilets, and premium complimentary hot meals onboard.

ROUTE TIMELINE

- Delhi**
Departure - 08:00 AM (Origin)
- Aligarh Junction**
Stop - 2 min halt
- Kanpur Central**
Stop - 2 min halt
- Prayagraj Junction**
Stop - 2 min halt
- Varanasi**
Arrival - 08:00m (Destination)

ON-BOARD COMFORTS

- Pantry Car
- AC Comfort
- Power Plugs
- Wi-Fi Link

GEMINI JOURNEY INSIGHTS

Here is your travel insight summary for the Vande Bharat Express (22436) from New Delhi to Varanasi:

Punctuality & Performance As a high-priority train, it maintains an exceptional punctuality record

Fig. 1.9 Train Search Page

— Back to Search

1 Journey Review

2 Passenger Details

3 Secure Payment

★ 4.9 RATED SERVICE

Patna Vande Bharat Express
Train No: 22346 • Speed-Link
New Delhi → Patna

DATE OF JOURNEY: 2026-08-02

Fare Summary

RECEIPT

Base Ticket Fares (1 pax)	₹1780
Reservation Fee	₹60
Superfast Surcharges	₹45
IRCTC Convenience Flat Fee	₹15
Total Payable Amount	₹1900

Travel Insurance Included
Complimentary ₹10 Lakhs accident coverage included automatically.

Proceed to Payment →

SSL Secure Encrypted Transaction

Ticket Preferences

Class of Ticket	Number of Passengers	Preferred Coach
CC	1 Passenger	Coach C2

Passenger Details

SEAT 1 SELECTION

Full Name	Age	Gender
Aarshi Mam	26	Female

Birth (Seat) Preference

Side Lower

Fig. 1.10 Passenger Detail Page

← BACK

SECURE PAYMENT

Credit/Debit Card

UPI Payment

PhonePe

Razorpay

Amazon Pay

UPI-Pai

ENTER UPI ADDRESS (VPA)

kirtianurag@ybl

Pay Now

ALREADY WITH VERCEL & STRIPE

ORDER SUMMARY

RECEIPT

Receipt

Train Route (22348) → (MIN-2AR)

Travel Class AC CHAIR CAR

Booking Status BOOKING

Base Ticket Fare ₹1780.00

Superfast Fee ₹30.00

Convenience Charge ₹20.00

Total Payable ₹1830.00

SSL Encrypted ✓ IRCTC Verified

Fig. 1.11 Payment Page

← Back to Dashboard

✓ BOOKING REFERENCE VERIFIED

DIGITAL BOARDING HUB

ELITE RAILWAY

NDLS

New Delhi

→

PNBE

Patna

BOOKING CONFIRMATION

#3969456

DATE & TIME

2020-08-09 • 08:30 AM (ETD)

TRAIN

12394 Sampurna Kranti Express (1A)

PNR REFERENCE

3969456540

SEAT / COACH

Coach C1, Seat WL 2

PASSENGERS (1)

1. Aarshi Mam (26 yrs) WL 2

E-TICKET

Security details provide passenger validation. Ticketing terms are secure and encrypted under Ministry of Railways (IRCTC).

▲ PASSENGER TRAVEL REGULATIONS

Please carry a valid original identity proof (Aadhaar Card, Passport, or Voter ID) during your journey. Passengers are advised to arrive at the boarding terminal at least 30 minutes prior to departure. This is an electronically generated ticket; physical copies are not mandatory under Digital India compliance guidelines.

RAILWAY GENERAL INQUIRY: 139

RAILWAY SECURITY HELP: 182

IRCTC DIGITAL SUPPORT: eickets@irctc.com

Fig. 1.12 Final Ticket

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Link: <https://vitejs.dev/>
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Link: <https://www.irctc.co.in/>

Deployment Link: <https://railway-ai-project.vercel.app/>